
CS 677: Soft Computing

Course Code	CS 677
Course Name	Soft Computing
L-T-P-C	2-0.5-0.5-3
Prerequisites	Knowledge of Probability, Statistics and Optimization, C/C++/MATLAB.
Intended for	M.S./Ph.D
Distribution	Elective
Approval	4th Senate (May 2013)

Course Description

Introduction to soft computing and its applications. Biological neural network, artificial neural networks and applications, mathematical foundations and learning mechanisms, perceptron learning, radial basis function networks, feed forward neural network, competitive learning, self organizing neural network, support vector machine, Application of to Financial Markets, weather prediction. Fuzzy systems and applications: Introduction to fuzzy sets and fuzzy logic systems, fuzzy sets, fuzzy reasoning, fuzzy inference systems, fuzzy clustering, applications of fuzzy systems, neuro-fuzzy modeling and control.

Text Books

1. Simon Haykin, **Neural Networks**, Prentice Hall, 1998.
2. Timothy J Ross, **Fuzzy Logic with Engineering Applications**, Wiley, 2009.
3. J.S.R. Jang, C.T. Sun, E. Mizutani, **Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence**, Prentice Hall, 1996

References

1. Mohamad H. Hassoun, **Fundamentals of Artificial Neural Networks**, MIT Press, 1995.
2. Bart Kosko, **Neural Networks and Fuzzy Systems**, Prentice Hall of India Learning, 2009.
3. M. Friedman and Abraham Kandal, **Introduction to Pattern Recognition- Statistical, Structural, Neural and Fuzzy Logic Approaches**, World Scientific, 2005.